Students Learning Journey Summarization System

Transforming Raw Data into Actionable Insights

Analytics
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AI-powered
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Education
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Final Presentation • May 10, 2026

Background

The project addresses several critical gaps in the utilization of educational data :

Problem & Fragmented Learning Data

The rapid digitization of education has resulted in the continuous generation of diverse learning data,. Despite this abundance, most educational data remains fragmented and underutilized, offering limited value in guiding instruction or supporting student reflection. Learning data from assessments, reflections, and activities often remains fragmented, making it hard for educators to extract timely insights. An intelligent system that automatically integrates and summarizes this data can provide clear, actionable understanding to support better teaching and learning outcomes.

Observation

This made me wonder: *What if AI could automatically analyze this information and summarize progress, strengths, and weaknesses for each learner?*

That curiosity led to my FYP — developing an AI-powered Educational Data Analytics Platform that uses Large Language Models (LLMs) that uses genAI Transformers and Pytorch to summarize learning data and generate clear, visual, and personalized reports for students and educators.

Research Gap: Transforming raw educational data into actionable insights through unified, automated systems

Problem Statement

Data Analysis
We have plenty of learning data — but very little understanding.

Actionable Insights
Current systems fail to provide meaningful analysis

Despite the abundance of data within different platforms, a substantial disconnect exists between data collection and its effective utilization:

Limited Actionable Insights

Students’ activities, assessments, and reflections are often scattered across multiple platforms or formats. This fragmentation prevents educators and students from obtaining an adaptive support and holistic evaluation.

Manual Analysis

Instructors resort to manually downloading and interpreting multiple datasets, a time-consuming process error-prone and inaccessible to those without data-analysis training.

Lack of Personalized Feedback

Students receive little in the way of meaningful explanations, progress trends, or personalized feedback from existing systems.

Underutilized Pedagogical Data

This fundamental mismatch leads to vast amounts of collected data remaining pedagogically underutilized, hindering opportunities for improved teaching and learning.

Research Objectives

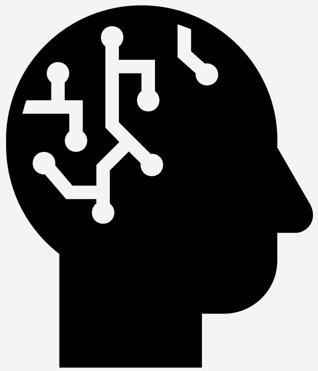
The study aims to achieve the following five main objectives:

1. Automated Summarization & Insight Generation

The objective of this project is to develop an automated data analysis system that summarizes student learning performance using both numerical results and text-based feedback, providing clear insights into progress trends, strengths, and improvement areas.

2. Interactive Visualization Components

The objective of this project is to design an automated visualization system that transforms uploaded learning data into interactive dashboards, generating appropriate charts or text-based graphics to highlight performance patterns and key insights.



Evaluation Metrics (Objective- 1)

Aspect	How It Is Achieved	Metric	Measurement Method & Justification
Summarization Accuracy	LLM generates summaries from uploaded PDF/CSV/Excel data.	**BLEU / ROUGE Score**	Compare model-generated summaries with manually written benchmarks. These NLP metrics are standard for evaluating content overlap and linguistic accuracy in summarization systems. Target: BLEU ≥ 0.75 or ROUGE-L ≥ 0.80.
Content Relevance	Summaries filtered for key learning indicators.	**Relevance Score (1–5)**	Educator experts or collaborator rate how well generated summaries reflect actual learning outcomes. This subjective metric complements BLEU/ROUGE by validating contextual relevance.

Evaluation Metrics (Objective- 1)

Aspect	How It Is Achieved	Metric	Measurement Method & Justification
Response Efficiency	Optimized inference and prompt tuning.	**Average Processing Time**	Time from upload to summary generation (< 30 seconds). Ensures practical performance for real-time classroom use.
User Satisfaction	Feedback collected from educators/students.	**Satisfaction Rating (%)**	Collaborator & students survey (Likert scale 1–5 or Yes/No). Quantifies perceived usefulness and accuracy. A satisfaction rate ≥ 80 % indicates positive acceptance.
Readability	NLP post-processing ensures coherent phrasing.	**Flesch–Kincaid Index**	Measures text clarity and comprehension level. Target: readable by a general academic audience (grade 8–10).

Evaluation Metrics (Objective- 2)

Aspect	How It Is Achieved	Metric	Measurement Method & Justification
Numerical Data Visualization	Performance and progress trends are shown using interactive **bar, line, and pie charts** created with Plotly.	**Usability Score / SUS**	Measured through collaborator surveys and the **System Usability Scale (SUS)**. This approach quantifies how easily users can interpret and interact with the charts. A usability score of ≥ 80 % or SUS ≥ 70 indicates strong acceptance.
Text-Based Visualization	**Word-frequency charts** and keyword highlights reveal learning themes in textual data.	**Interpretability Rating**	Educators and students answer “Was the visualization easy to understand?” on a 1–5 Likert scale. This human-centered measure ensures the visual output communicates insights clearly rather than just displaying data.

Evaluation Metrics (Objective- 2)

Aspect	How It Is Achieved	Metric	Measurement Method & Justification
Performance	Optimized chart rendering and caching for fast response times.	**Load Time**	Must render in **< 5 seconds** (observed locally).

Problem Statement	Addressed by Objective	Explanation
Limited Actionable Insights	✅ **Objective 1**	The genAI LLM-driven summarization module automatically interprets student data to generate clear, actionable insights that help identify learning trends and challenges.
Manual Analysis	✅ **Objective 1**	Automating the summarization process removes the need for teachers to manually review multiple datasets, saving time and reducing errors.
Lack of Personalized Feedback	✅ **Objective 1**	The generated summaries include personalized reflections on strengths, weaknesses, and progress, directly addressing the lack of meaningful, individualized feedback.
Underutilized Pedagogical Data	✅ **Objective 2**	The interactive visual dashboards transform raw educational data into interpretable visuals, ensuring data is effectively used to support teaching and learning.

Project Impact & Benefits

This project significantly contributes to the vision of data-driven, personalized learning environments by transforming raw educational data into actionable insights.

Cabral et al. (2025)

- Adopted AI-powered Learning Analytics Dashboards (LADs) to collect and analyze LMS data (57% of studies) for summarizing learning behavior using predictive and descriptive analytics.
- Chosen for its ability to give real-time, personalized insights into student performance and engagement — engagement — aligning with the system's goal of summarizing learning journeys effectively.
- About **72 percent** of AI learning dashboards mainly focus on showing data and making predictions, around **18 percent**—actually provide personalized feedback.

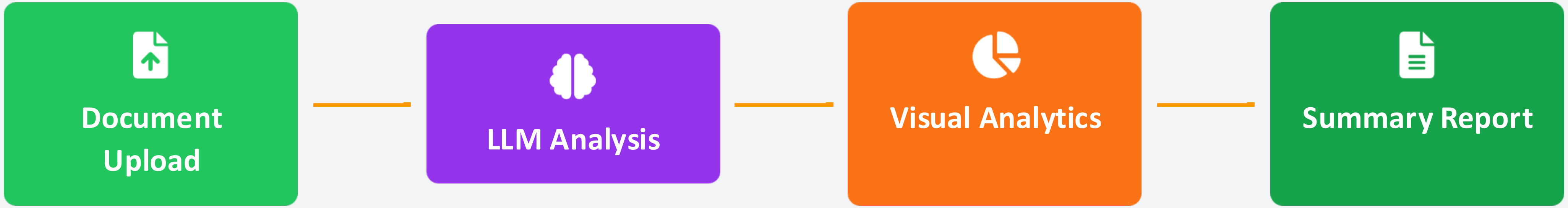
Oprea and Bara (2025)

- Over **9,598 educational LLM studies (2019–2024)** show exponential growth in AI-driven learning research, yet **less than 5%** address *personalized learning summarization*.
- Adaptive feedback and summarization tools can improve learning efficiency by **up to 37%**, according to educational analytics benchmarks
- Among **25,381 education-related terms**, "learning" (4,828), "research" (4,438), and "training" (3,027) dominated — confirming LLMs' strong alignment with adaptive and data-driven learning models.

Key Impact: Both studies emphasize the urgent need for **interpretable, LLM-driven, and multimodal educational analytics systems, systems**, directly supporting the development of your platform as a unified solution that transforms LMS data into actionable, personalized learning insights.

Proposed Solution Overview

Our system addresses the data-insight gap by introducing an LLM-powered framework that automatically analyzes educational documents educational documents and generates comprehensive summary reports with visual analytics.



Document Upload

A web form interface that allows users to upload files through a simple file selector component.

LLM-Powered Analysis

Leverages large language models to automatically analyze uploaded educational documents and generate context-aware summaries and insights.

Visual Analytics

Creates interactive data visualizations using Plotly and Streamlit to present learning trends, performance patterns, and key insights in an accessible format.

Comprehensive Reports

Generates downloadable analytical reports combining text-based insights and visual performance metrics for both educators and learners.

System Workflow Process

The Student Learning Overview System follows a structured five-stage workflow, transforming raw educational data into actionable insights through a clear, automated process.

1. Data Ingestion

Users upload structured and unstructured educational files (CSV, Excel, PDF,PNG).

2. Preprocessing

System cleans, harmonizes, and encodes data for multimodal analysis.

3. Visualization

Interactive charts and performance trends are generated using Plotly and Streamlit

4. Summarization

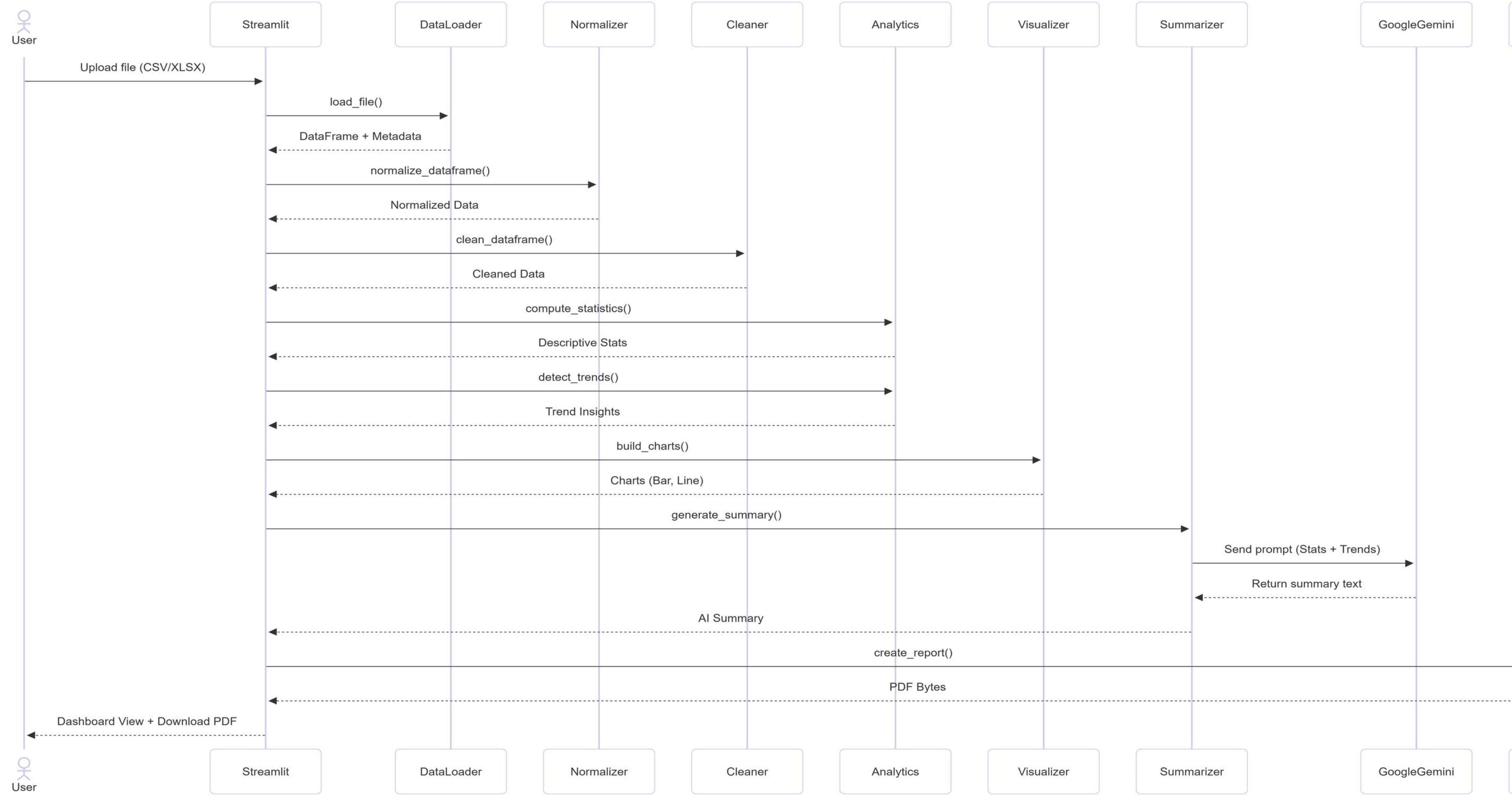
LLM-based summarization is applied to produce contextual insights and learning summaries.

5. Report Generation

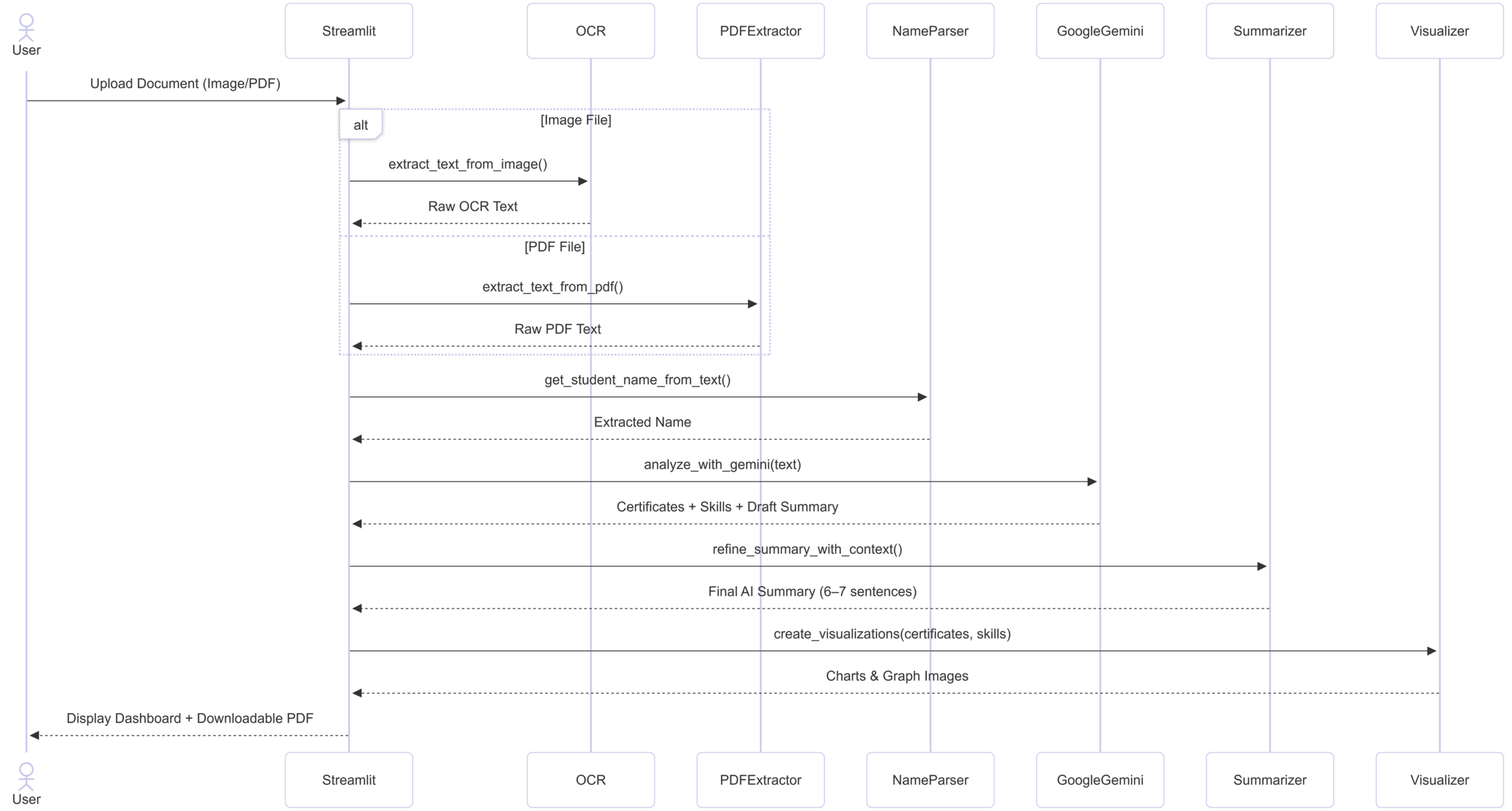
Results are compiled into a downloadable analytical report combining text summaries and visual analytics.

Key Benefit: This structured workflow ensures efficiency, scalability, and effective integration of diverse data types, transforming raw educational data into actionable insights.

Structured Data Processing Pipeline



Unstructured Data Processing Pipeline



System architecture diagram showing structured vs. unstructured data paths

7-step processing pipeline:

Input Stage (data_loader.py) – File type detection, OCR extraction for images, PDF parsing

Normalization (normalizer.py) – Standardizes 50+ column name variations to Score/Maximum

Cleaning (data_cleaning.py) – Removes duplicates, NaN values

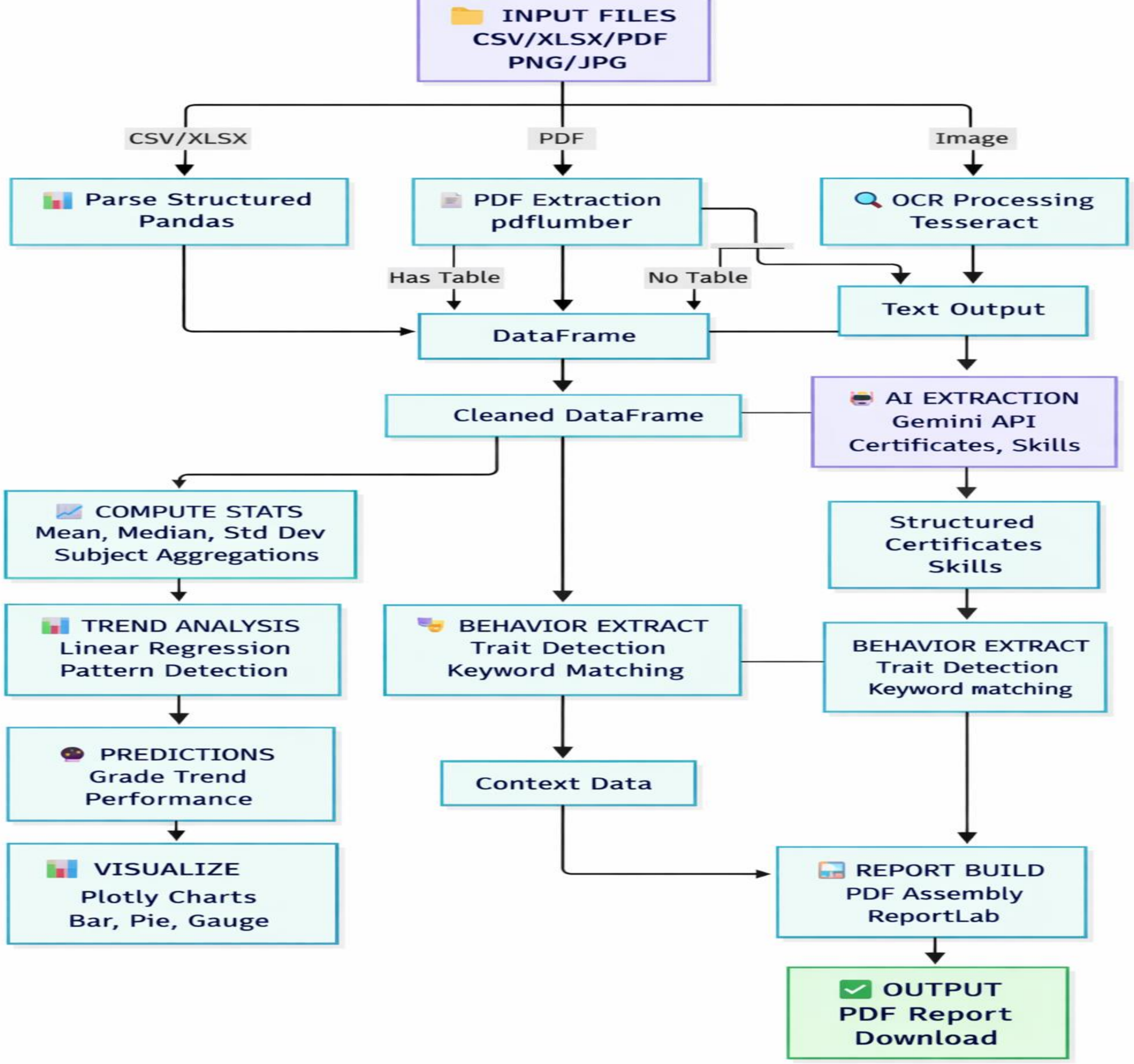
Analytics (analytics_statistics.py) – Computes mean, median, std dev per subject

Extraction (behaviour_extractor.py, text_info_extractor.py) – Extracts traits and student metadata

Visualization (chart.py) – Renders Plotly charts (Bar, Pie, Gauge)

Final Output (summarizer.py + download.py) – PDF generation and dashboard display

Key Detail: All modules feed into a single stats dictionary that flows through the system


INPUT FILES CSV/XLSX/PDF PNG/JPG
CSV/XLSX
PDF
Image
Parse Structured Pandas
PDF Extraction pdflumber
OCR Processing Tesseract
Has Table
No Table
Text Output
DataFrame
Cleaned DataFrame
AI EXTRACTION Gemini API Certificates, Skills
COMPUTE STATS Mean, Median, Std Dev Subject Aggregations
Structured Certificates Skills
TREND ANALYSIS Linear Regression Pattern Detection
BEHAVIOR EXTRACT Trait Detection Keyword Matching
BEHAVIOR EXTRACT Trait Detection Keyword matching
PREDICTIONS Grade Trend Performance
Context Data
VISUALIZE Plotly Charts Bar, Pie, Gauge
REPORT BUILD PDF Assembly ReportLab
OUTPUT PDF Report Download

Unstructured Certificate Processing:

Input: Scanned PDF,CSV,XLSX,XLS & Image certificates

Gemini receives prompt: "Extract certificate name, date, organization, skills. Return JSON.“

Output: Structured data (certificates dict with name, date, organization, skills).

Numeric Data Summary Generation:

Input: CSV,Image,PDF,XLSX,XLS with grades (example: Average=78.5, Highest=92, Behavior=hardworking)

Gemini receives prompt: "Write 6-7 sentence professional summary with strengths, development areas, recommendations“

Output: "Nur Aisyah Binti Rahman, a Form 5 Secondary student from Johor, Malaysia, consistently demonstrates strong academic engagement. Her academic performance shows a particular strength in History, with Mathematics identified as an area for improvement. ……”

Graceful Degradation:

If Gemini API unavailable → Falls back to regex extraction

If rate limit hit → Uses template-based summary

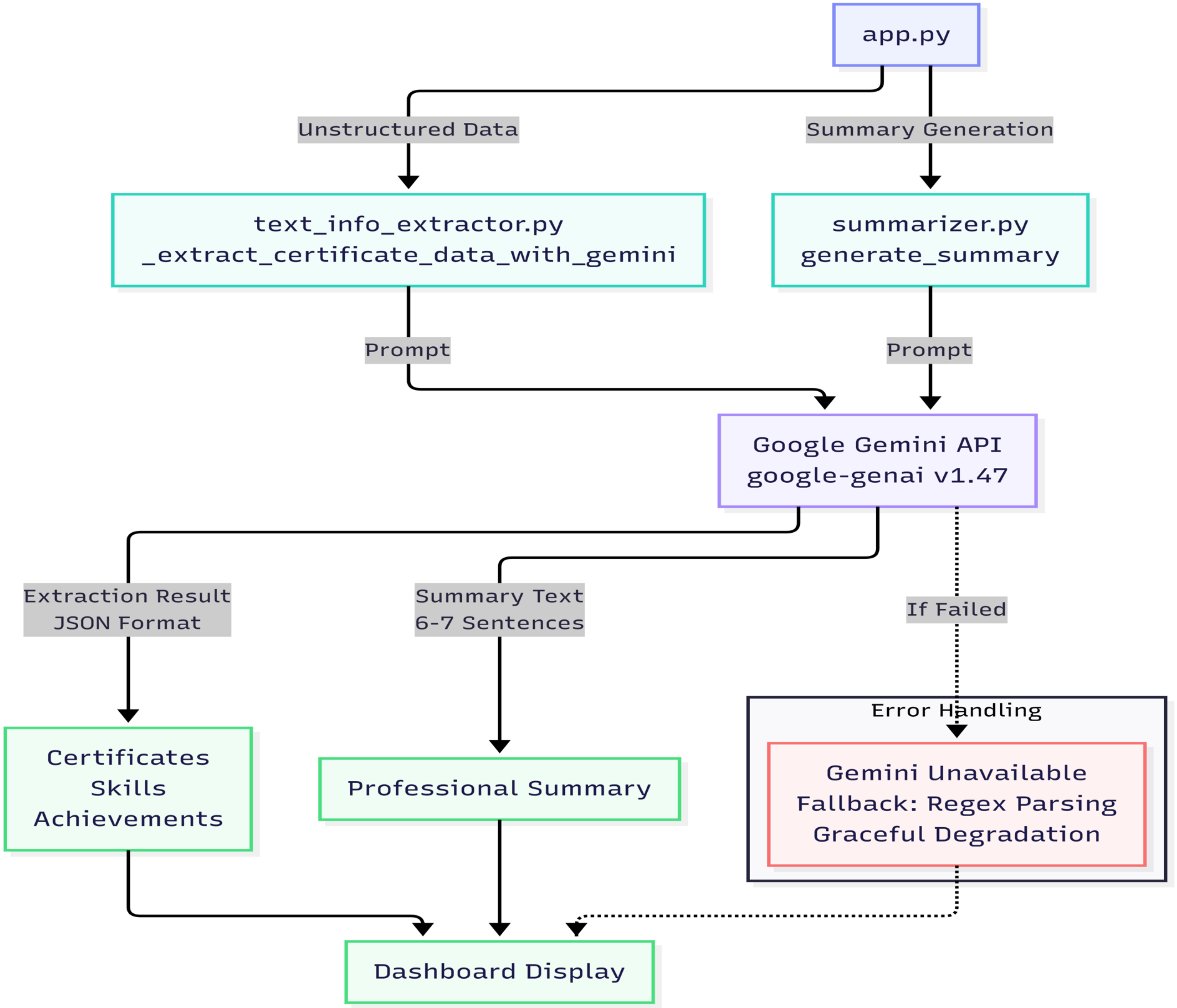
App remains functional without AI

AI Integration Points:

text_info_extractor.py – Calls Gemini for unstructured document parsing

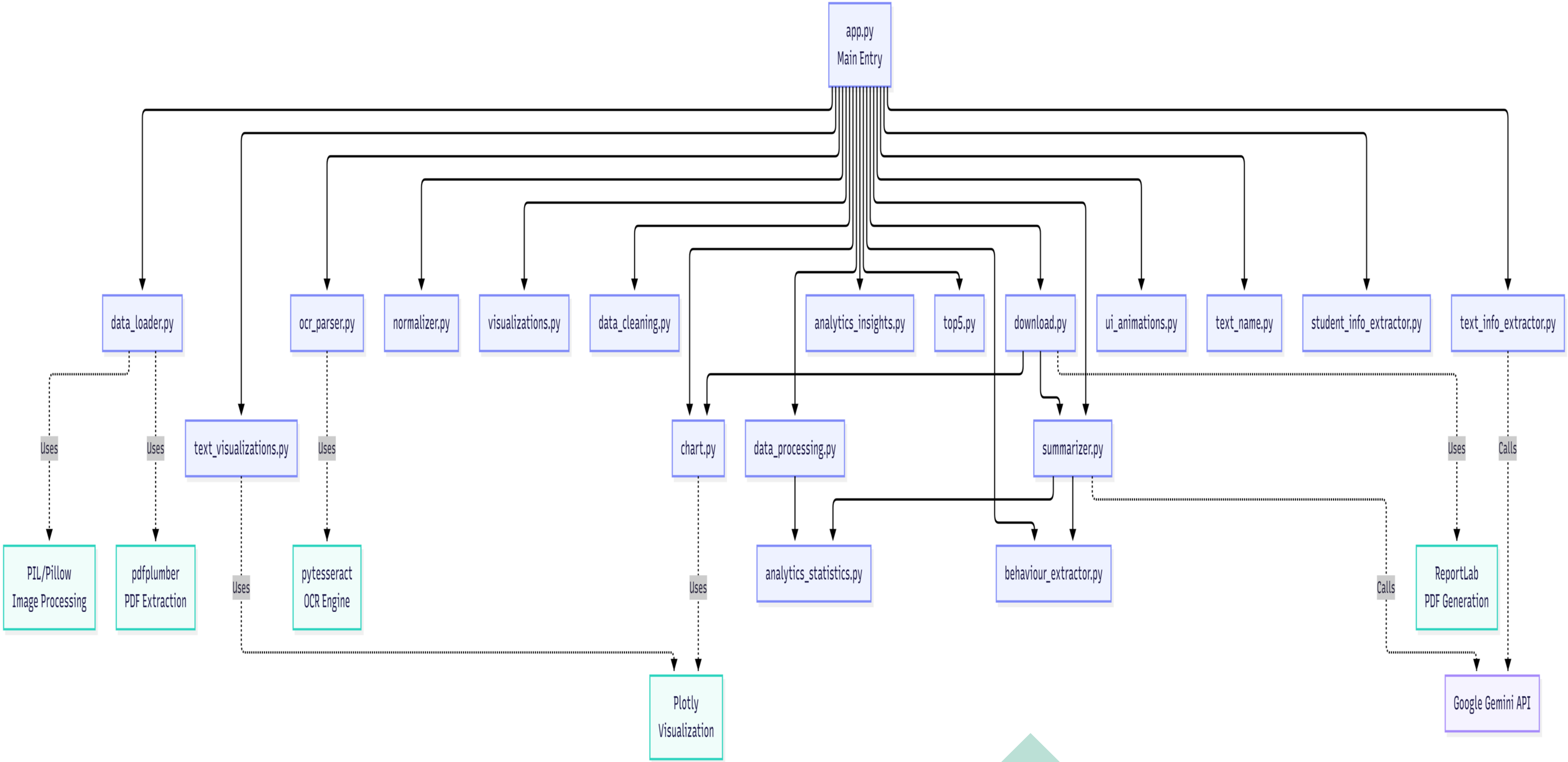
summarizer.py– Calls Gemini for professional summary generation

Both have exponential backoff retry logic and graceful error handling


app.py
Unstructured Data
Summary Generation
text_info_extractor.py
_extract_certificate_data_with_gemini
summarizer.py
generate_summary
Prompt
Prompt
Google Gemini API
google-genai v1.47
Extraction Result
JSON Format
Summary Text
6-7 Sentences
If Failed
Error Handling
Gemini Unavailable
Fallback: Regex Parsing
Graceful Degradation
Certificates
Skills
Achievements
Professional Summary
Dashboard Display

Complete data flow showing 8 tiers of modules:

Tier	Module	Function	Integration
Tier 1	data_loader, ocr_parser	Input ingestion	Routes file to processor
Tier 2	normalizer, data_cleaning, data_processing	Standardization	All data passes through before analysis
Tier 3	analytics_statistics, analytics_insights, behaviour_extractor	Analysis	Creates stats dictionary
Tier 4	text_info_extractor, student_info_extractor, text_name	Extraction	Adds metadata to stats dict
Tier 5	summarizer	Summarization	**AI call with full context**
Tier 6	chart, ui_animations	Visualization	Generates display components
Tier 7	download	Report assembly	Converts all components to PDF
Tier 8	app.py	Orchestration	Calls all tiers in sequence



app.py
Main Entry
data_loader.py
ocr_parser.py
normalizer.py
visualizations.py
data_cleaning.py
analytics_insights.py
top5.py
download.py
ui_animations.py
text_name.py
student_info_extractor.py
text_info_extractor.py
text_visualizations.py
chart.py
data_processing.py
summarizer.py
Uses
Uses
Uses
Uses
Calls
PIL/Pillow
Image Processing
pdfplumber
PDF Extraction
pytesseract
OCR Engine
analytics_statistics.py
behaviour_extractor.py
ReportLab
PDF Generation
Uses
Uses
Calls
Plotly
Visualization
Google Gemini API

Functional Requirements

The system is designed with key functional requirements to ensure effectiveness and usability:

File Upload & Input Handling

- Supports PDF, CSV, Excel & Image formats
- Validates file format and size upt o 200 MB
- Merges multiple documents into single dataset

Data Ingestion & Preprocessing

- Extracts content from structured and and unstructured data
- Cleans and normalizes structured data
- Applies NLP preprocessing to unstructured text

LLM-Based Summarization

- Generates concise, context-aware summaries
- Identifies learning progress and strengths/weaknesses
- Creates integrated analytical reports

Data Visualization

- Creates interactive visualizations for grades and trends
- Enables user interaction with visuals
- Displays correlations between engagement and performance

Report Generation & Export

- Compiles text and graphs into downloadable reports
- Includes executive summary and key findings
- Generates model-based recommendations

Dashboard Interface

- Provides intuitive web-based dashboard and download option
- Displays system feedback messages
- Supports user-friendly data interaction

Non-Functional Requirements

Beyond core functionalities, the system adheres to several non-functional requirements to ensure quality, reliability, and user satisfaction:

Performance

- Processes documents within 60 seconds
- Renders visualizations in under 5 seconds

Usability

- Simple user experience for all users
- Clear, structured reports

Reliability

- Handles simultaneous uploads
- Validates partial data files

Scalability

- Handle simultaneous data uploads
- Maintain consistent performance

Security

- Secure data handling
- No long-term persistence of uploaded content beyond processing

Maintainability

- Modular architecture
- Comprehensive module documentation

Accuracy & Quality

- 75%-80% accuracy rate for summaries
- Visualizations match data analysis

Availability

- 99% uptime during operational hours
- Fallback explanations for errors

Technology Stack & Implementation

The project leverages a robust set of tools and frameworks, with skill growth observed across different development stages.

Frontend

- HTML/CSS
- Streamlit
- Scipy

Backend

- Python
- Altair
- Google Gemini API

File Processing

- PDFPlumber
- Tesseract OCR
- NLP Libraries

AI Integration

- Google-genai
- Docker
- Fallback Regex

Skill Growth Across Development

Development Phase
Progressive skill acquisition across planning, implementation, and testing phases

LLM Integration
Enhanced understanding of educational data through LLM capabilities

Implementation Approach

Modular Architecture
Robust, scalable design with clear separation of concerns

Cloud Integration
Secure, scalable cloud infrastructure for data processing

Tool / Framework	Purpose	FYP 1 (Basic Prototype)	FYP 2 (Advanced System)	Skill Growth
Streamlit	Dashboard & Interface Design	Build a simple interface that allows users to upload files (CSV/XLSX) and view extracted data with basic charts (bar, pie) and text summaries.	Add user login with authentication, session management for multi-user support, real-time data updates, dark/light theme toggle, and performance metrics dashboard.	Beginner → Advanced
Google Gemini API	AI-Powered Text Summarization & Extraction	Use rule-based text extraction (regex patterns) and template-based summaries to simulate AI intelligence without calling external APIs.	Integrate Gemini API for intelligent certificate parsing, skill extraction from unstructured documents, and dynamic professional summary generation with prompt engineering.	Introductory → Advanced
Pandas & NumPy	Data Handling & Processing	Load and clean CSV/Excel files, display simple statistics (averages, totals, min/max scores) and basic subject-level aggregations.	Extend analysis with advanced predictive insights using linear regression for trend detection, multi-level grouping, statistical hypothesis testing, and support for image file uploads with OCR.	Intermediate → Advanced
PDFPlumber & OpenPyXL	File Reading & Extraction	Extract text and tables from uploaded PDFs and Excel files, parse metadata (student name, dates), and display raw extracted content on dashboard.	Enhance extraction accuracy using Tesseract OCR for scanned documents, implement intelligent column mapping, handle structured data transformation, and build fallback extraction pipelines.	Intermediate → Advanced

Tool / Framework	Purpose	FYP 1 (Basic Prototype)	FYP 2 (Advanced System)	Skill Growth
Tesseract OCR & pytesseract	Image-to-Text Extraction	Placeholder: Display message if OCR attempted but not implemented.	Implement full OCR pipeline for scanned certificates and images, convert PNG/JPG files to structured DataFrames, and integrate with Gemini for intelligent text understanding.	Beginner → Intermediate
ReportLab & python-docx	PDF Report Generation	Generate basic PDF with student name, scores, and simple tables; static layout.	Create professional, multi-page reports with embedded charts, styled tables, footer/header, dynamic content based on student profile, and batch report generation for multiple students.	Beginner → Intermediate
Python Regex & NLP Patterns	Text Parsing & Information Extraction	Use regex to extract student names, identify subjects, and parse basic metadata from text or file headers.	Build advanced text parsing pipeline with fuzzy matching for subject names (handle variations like "Sains" = "Science"), behavior trait extraction using keyword hierarchies, and confidence scoring for extractions.	Beginner → Intermediate
Plotly & Altair	Data Visualization	Create simple bar (subject performance), line (score trends), and pie (score distribution) charts to visualize student performance data.	Develop interactive, animated dashboards with AI-driven visual insights, gauge charts for performance levels, dynamic filtering by subject/date, and export-ready report visualizations.	Beginner → Advanced

Tool / Framework	Purpose	FYP 1 (Basic Prototype)	FYP 2 (Advanced System)	Skill Growth
Environment Management (.env)	Configuration & API Key Handling	Store API keys and config in hardcoded strings or simple config files.	Use `.env` files with `python-dotenv` for secure credential management, implement environment-specific configurations (dev/staging/prod), and add validation for required variables at startup.	Beginner → Beginner
Docker & Containerization	Deployment & Reproducibility	Create a Dockerfile with basic Python 3.11 setup and requirements.txt.	Optimize Docker image with multi-stage builds, add health checks, implement volume mounts for persistence, and set up registry deployment.	Intermediate → Advanced

Collaboration Letter

Acceptance of Collaboration

I hereby acknowledge that I accept this proposal for collaboration with the Faculty of Computer Science & Information Technology, Universiti Malaya pertaining to the research project entitled **Student Learning Journey Summarization System.**

Collaborator signature

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Ms Foo Yet San,
Research Associate,
MICSS Educational Research and Development Unit
United Chinese School Committees' Association of Malaysia

Date: 2/1/2026

ANALYSIS SUMMARY

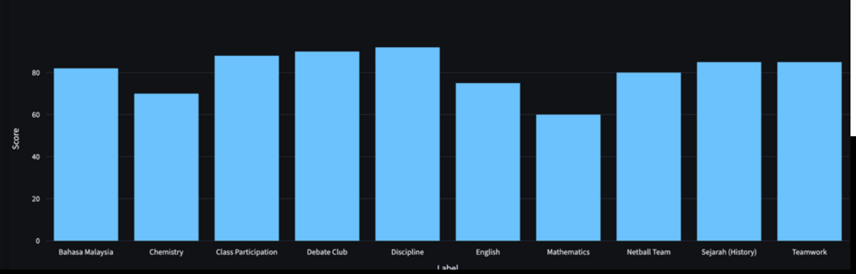
Nur Aisyah Binti Rahman, a Form 5 Secondary student from Johor, Malaysia, consistently demonstrates strong academic engagement. Her academic performance shows a particular strength in History, with Mathematics identified as an area for improvement. She achieved an overall average score of 80.7 out of a possible 100.0. Beyond academics, Nur Aisyah is actively involved in the Netball Team and Debate Club, reflecting a balanced personal development.

Result & Discussion (Objective-1)

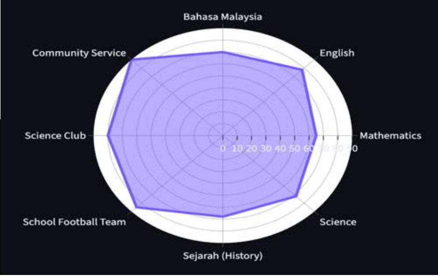
The summarization system provides information on **student performance**, making it easier for educators to interpret large datasets quickly.

By highlighting both **strengths and areas needing improvement**, the system supports **targeted interventions** and personalized learning strategies.

The inclusion of extracurricular involvement alongside academic results demonstrates a **holistic view of student development**, encouraging balanced growth.



Result & Discussion (objective-2)



Graph	Best for	What it shows	When to use
Bar	Comparison across categories	Relative scores or counts per category	Compare subjects or activities side-by-side
Line	Trends over time	Trajectory and rate of change	Track progress across assessments or terms
Area	Cumulative contribution	Magnitude plus trend with emphasis on totals	Show cumulative achievement or stacked components
Scatter	Relationship and spread	Correlation and outliers between two metrics	Explore score vs attendance or effort vs outcome
Pie	Proportional breakdown	Share of total by category	Show composition of time, credits, or activity mix
Spider	Multidimensional balance	Relative strengths across many axes	Present holistic profile across competencies

Project Impact & Benefits

This project significantly contributes to the vision of data-driven, personalized learning environments by transforming raw educational data into actionable insights.

Impact on Educators

- Empowered with richer, more actionable insights to support adaptive and personalized instruction
- Access to integrated analysis of both structured data (grades, attendance) and unstructured data (essays, reflections)
- Time-saving automation of data analysis that was previously manual and error-prone

Impact on Students

- Receives personalized feedback that supports reflective reflective and self-directed learning
- Access to comprehensive summary reports highlighting progress, strengths, and areas for improvement
- Transforms passive data into pedagogically valuable insight for improved learning outcomes

Key Impact: By automating the analysis of educational data and providing clear, evidence-based feedback, the system bridges the gap between data collection and effective teaching/learning.

Future Work & Conclusions

Conclusions

- Provides an intelligent bridge between student performance data and personalized insights through AI-driven summarization.
- Converts complex academic information into clear, interpretable learning pathways.
- Uses Large Language Models (LLMs) to enhance comprehension, reflection, and decision-making in education.
- Promotes a data-informed learning culture, enabling students and educators to monitor progress more effectively.

Impact

The project enables personalized academic reflection for students by grounding insights in real performance evidence, while at the same time assisting educators with faster progress analysis and more targeted feedback delivery.

Project Completion

December 07, 2025